Financial metrics and Gate B tests

Version 2. metrics_version v2, gate_version v2. One formula per metric, the gate, its calibration, and what changed from v1.

> **Notation.** r, b and c are one day's return on the strategy, the benchmark and cash. mean and stdev are the average and sample standard deviation. The benchmark is the NIFTY 500 price index, with no dividends, which flatters every relative figure. **A run's cost and cash rate come only from its card, as confirmed at Gate A.** If a card names no cash rate, Sharpe is over zero and the page says so. All code is in ros/metrics/core.py.

Part 1. The metrics

Shown on every run, in the Core performance table on the Results tab and the Core metrics sheet in Excel.

Metric	Formula	How to read it
CAGR	$(V_{end} / V_{start})^{1/years} - 1$, on calendar days / 365.25	Steady yearly growth rate
Volatility	stdev(r) x sqrt(252)	Lower is smoother
Sharpe	mean(r - c) / stdev(r - c) x sqrt(252)	Return above cash per unit of risk. **The only Sharpe**
Sortino	mean(r - c) x 252 / (sqrt(mean(min(r - c, 0)^2)) x sqrt(252))	As Sharpe, counting only falls as risk
Calmar	CAGR / max drawdown	Return per unit of worst fall
Max drawdown	max(1 - V_t / highest V so far), daily, positive	Worst fall from a high
Drawdown duration	Longest stretch below a previous high, in trading days	Longest wait to recover
95% CVaR	-mean(r on days at or below its 5th percentile)	Average loss on the worst days, positive
Skew	Sample skew of r	Negative: big falls outnumber big rises
CAGR without best days	CAGR after removing the 10 best days, same span	Did a few days make the result?
Beta	Cov(r - c, b - c) / Var(b - c)	Sensitivity to the index
Tracking error	stdev(r - b) x sqrt(252)	How far it strays from the index
Information ratio	mean(r - b) x 252 / tracking error	Return over the index per unit of straying
Up / down capture	mean monthly r / mean monthly b, in months b rose / fell	Want high up, low down
Rolling periods behind	Share of rolling 252-day periods with compounded r below b	How often an investor trailed

Costs, capacity and timing

Metric	Formula	How to read it
Gross and net	Daily returns are at the card's cost, so they are net. Gross = r + two-way turnover x card bp / 2 / 10,000	A round trip is one buy and one sell, so each unit of two-way turnover pays half. At 0bp on the card there is one number: gross equals net
Two-way turnover	Sum of |change in weight| per trade, / years	1.0 means trading capital once a year
Capacity	|change in weight| x AUM / average traded value of the 20 sessions before the trade	Median, 95th percentile, maximum and names over the limit. Needs AUM
Post-publication	The same metrics from the paper's date onward. A bare year means its end	Did it survive being published?
Period returns	Compounded r by calendar year, financial year (April to March), month, sub-period	Where the return came from

Strategy-specific rows go in a separate Diagnostics section. Each run can add its own further rows to the tables at the bottom of the Results tab, so those still differ by run. Kurtosis is kept only as an input to the deflated Sharpe, which stays in the code.

Calculated elsewhere

The Excel tearsheet (universal_backtester/excel_tearsheet.py and tearsheet.py) stays, because the Excel calculation is still needed. It holds Sterling, Treynor, Omega, Gain to Pain, tail ratio, Jensen's alpha, extreme capture and the rest. It is **not** part of the one-definition rule: it uses monthly returns and its own rates, so its Sharpe will not match the dashboard's. A geometric Sharpe survives in ros/metrics/legacy_v1.py only to compare with a paper's printed figure.

Part 2. Gate B v2

Status is PASS or FAIL. The decision stays PENDING until a named person records one. Every hard block that can be run must pass, and net edge and the bootstrap can always be run. A test with nothing to run on, such as no book, AUM or mandate limits, shows as **not run**: it is not a pass, it is listed beside the status, and it does not hold the gate up. Every number is at the cost on the run's card, and no other cost is applied. "Evidence supports" remains Claude's written read.

Hard blocks

Test	Measure	Pass mark	Status now
Mandate	Actual positions against config/mandate.yaml. A card cannot override it; an override needs a named approval with a timestamp	No breaches	Not run: no limits supplied
Capacity	95th percentile of trade size / 20-day traded value at the fund's AUM	≤ 10%	Not run: no AUM supplied
Data integrity	Look-ahead checks, the audit of rules, and share of index members with a price. Unpriced names are listed	All pass, coverage ≥ 90%	Runs when the run's files have them
Edge	CAGR at the card's cost - benchmark CAGR	> 0	Always runs
Paired bootstrap	Share of 2,000 draws (mean block 21, seed 7) with $\text{Sharpe}_{\text{strategy}} - \text{Sharpe}_{\text{benchmark}} \le 0$	P ≤ 0.10	Always runs

Evidence scorecard: green, amber or red

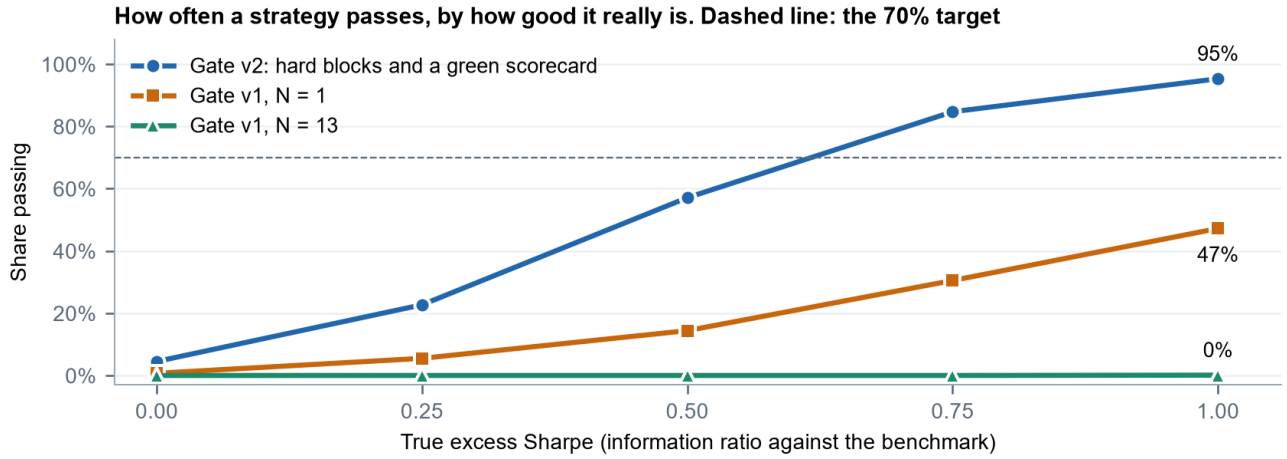
Test	Measure	Green
Sub-period stability	Sharpe gap to the benchmark in 4 equal windows. Called a walk-forward only if the card says the rules fit parameters, and then it skips the first 5 years	At least 50% of windows ahead, none below -1.00
Factor alpha	Regress r - c on market, Midcap 150, Smallcap 250, momentum, value, quality and low-vol, with HAC errors. t = alpha / its standard error. Each factor is also left out in turn	Alpha > 0 and t ≥ 2.0. One-sided: negative alpha can never go green
Book correlation	Largest correlation to the existing book, when a book file is supplied	≤ 0.80
Book IR gain	Best gain in the book's information ratio with a 5%, 10% or 20% sleeve	≥ 0.05
Post-publication	Edge from the paper's date onward, at the card's cost	> 0%

Amber bands are for display and were not simulated. The factor indices are backfilled before their launch dates and price-only. Alpha net of factor product costs needs those costs, which are not supplied yet.

> **The deflated Sharpe is out of the gate for now, by your ruling.** So nothing in Gate v2 tests for selection luck: trying many versions and reporting the best will not be caught. If it returns, its bar starts at the benchmark's Sharpe, also by your ruling, and the code and setting are still there (config/gate_thresholds.yaml, enabled: false). The figures below show the cost.

Part 3. Calibration

Set by simulation from 8,000 strategies, with no named strategy used: 12.7 years of daily returns, fat tails, volatility clustering, about 17% volatility and 0.8 correlation to the benchmark. The thresholds were eased at your request, as far as the 5% false-pass target allows. They were committed before any named strategy was scored on them.



Gate	True Sharpe 0	0.25	0.5	0.75	1.0	Factor only	Best of 20	Best of 100
v1, N = 1	0.7%	5.5%	14.4%	30.5%	47.2%	0.7%	10.1%	25.1%
v1, N = 13	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%
v2 status PASS (hard blocks)	6.9%	31.2%	63.2%	87.4%	95.7%	73.4%	72.5%	97.3%
v2 PASS and all green	4.5%	22.7%	57.1%	84.7%	95.3%	2.6%	59.6%	95.9%

"True Sharpe" is the true information ratio against the benchmark. Best of N is the best of N strategies with no skill, chosen in-sample. Scored with no book. Simulated returns are treated as already at their own cost. Capacity, mandate and data integrity are not modelled.

Target	Result
False pass at most 5% at true Sharpe 0	**Met:** 4.5% on PASS and all green. But 6.9% reach status PASS, above 5%, because the scorecard does not block
False pass at most 5% on pure-factor strategies	**Met on PASS and all green:** 2.6%. **Not on status alone:** 73.4% reach PASS with factor alpha red
False pass at most 5% on selection bias	**Not met:** 59.6% of the best of 20 and 95.9% of the best of 100 pass. This is the cost of removing the deflated Sharpe
Pass at least 70% at true Sharpe 1.0	**Met:** 95.3%

> **So read status and scorecard together.** PASS alone means only that the strategy beat the benchmark, at its card's cost, by more than noise explains. On a card set to 0bp, that says nothing about what trading would cost: read the turnover and the appendix cost sweep. Factor alpha is what separates skill from buying factors, and on the current rules it does not block.

Part 4. What changed from v1

v1	v2	Why
Deflated Sharpe blocks, at V = 1	Out of the gate for now	Your ruling. It failed almost everything: at N = 13, v1 passes 0.1% of strategies with a true excess Sharpe of 1.0
Worst window: absolute Sharpe ≥ 0.30	Sharpe gap to the benchmark, on the scorecard	A weak market should not fail a strategy that beat it
Bootstrap P ≤ 0.05, gross	P ≤ 0.10, at the card's cost	Your ruling to ease
Alpha |t| ≥ 2, blocking, on a narrower factor set	Alpha > 0 and t ≥ 2, on the scorecard, with size factors	v1 passed a strongly negative alpha. A missing size factor overstated alpha
Book tests advisory when no book	Not run when no book	Your ruling: a missing book does not hold the gate up
Turnover ≤ 400%, advisory	Reported only, plus capacity	Capacity is the practical limit. The cost is whatever the card states
Mandate judged with a flag on the card	Fund file the card cannot override	A card should not set its own limits
No edge test	Edge > 0 at the card's cost is a hard block	The strategy must beat the index
Cost and cash from the card, but 30bp and 6% also written into code and rules	Only the card. Sweep grids in config/ reach only an appendix	Your ruling: a cost set at Gate A is the cost charged
A list of 8 criteria	Hard blocks and a scorecard, PASS or FAIL	Separates what stops a strategy from what informs you